(*c*) What is underground powerhouse ? What are different types of underground powerhouse? 7

(*d*) Draw and explain a hydro-power station structure. 7

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320679(20)

BE (6th Semester)

Examination, April-May, 2016

(New Scheme)

Water Power Engineering

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Time Allowed : 3 hours *Maximum Marks* : 80

Minimum Pass Marks : 28

Note : (*i*) **All** questions are compulsory. Part (*a*) of all questions is compulsory. Solve any **two** parts from (*b*), (*c*) and (*d*) of each question.

(*ii*) The figures in the right-hand margin indicate marks.

Unit - I

1. (*a*) Define Peak load. 2

(*b*) What are the important considerations regarding site selection for a water-power schemes in India? 7

(*c*) Draw and explain typical line-diagram of a Hydro-Power Scheme, showing various components. 7

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(*d*) Write a short note on important conventional sources of energy in terms of Mass-Power Generation. 7

Unit - II

2. (*a*) Define Load factor. 2

(*b*) What is Mass curve ? How is it useful for calculating the storage capacity of a reservoir ? 7

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(*c*) A hydropower plant is proposed across the river site where net head of 30 m is available on the turbine. The river carries a sustain min. flow of 40 cumec in dry weather. Sufficient pondage is provided to supply daily peak load of demand with load factor of 75%. Assuming the plant efficiency of 60%, determine the max. generating capacity of the generator to be installed at the powerhouse. If the daily load pattern indicates 20 hr., average load 4 hr. peak load, determine the pondage to be provided to supply the daily demand. 7

(*d*) Write short notes on : $3\frac{1}{2} \times 2 = 7$

(*i*) Primary Power

(*ii*) Pondage Capacity

Unit - III

3. (*a*) Define forebay. 2

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(*b*) What are important components of a typical Hydropower Plant ? Draw a layout of medium head hydropower plant. 7

(*c*) Give general arrangement of run-of river plants and explain them with suitable diagrams. 7

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(*d*) Write a note on Diversion Canal Plant and also show its various layouts. 7

Unit - IV

4. (*a*) What is Water Conveyance ? 2

(*b*) Give classification of Penstocks with suitable sketches. 7

(*c*) A penstock with an internal radius of 0.6 m supplies water at a head equivalent to 18.5 kg/cm^2. There is a possibility of 20% increase in the pressure due to transient conditions. The design stress and the efficiency of the joint may be assumed to be 1030 kg/cm^2 and 85% respectively. Calculate the wall thickness of the penstock required. 7

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(*d*) What do you understand by 'Water-hammer' ? Derive an expression for rise of pressure in a thin rigid pipe of circular section in which flow is stopped by sudden close of valve. 7

Unit - V

5. (*a*) What is objective of power house planning? 2

(*b*) Write down various advantages and disadvantages of an underground power-house. 7